Final Project - Book Review Application

Estimated Time Needed: 2 hours

- In this final project, we will build a server-side online book review application and integrate it with a secure REST API server which will use authentication at the session level using JWT. You will then test your application using Promises callbacks or Async-Await functions.

Objectives:

After completing this lab, you will be able to:

1. Create APIs and perform CRUD operations on an Express server using Session & JWT authentication.
2. Use Async/Await or Promises with Axios in Node.js.
3. Create REST API endpoints and test them using Postman.

Set-up : Create application

1. Open a terminal window by using the menu in the editor: Terminal > New Terminal.
2. Change to your project folder, if you are not in the project folder already.

```
1
cd /home/project
```

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3. Please fork the Git repository that contains the starter code needed for this lab:

```
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https://github.com/ibm-developer-skills-network/expressBookReviews.git
```

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4. Clone your forked Git repository, if it doesn’t already exist.

```
1
[ ! -d 'expressBookReviews' ] && git clone https://github.com/<your Github username>/expressBookReviews.git
```

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5. Change to the directory **expressBookReviews/final_project/** directory to start working on the lab.

```
1
cd expressBookReviews/final_project/
```

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6. List the contents of this directory to see the artifacts for this lab.

```
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ls
```

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Understanding the server application

1. In the files explorer open the **expressBookReviews/final_project/** folder and view `index.js`having the below code:

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```

```
const express = require('express');
const jwt = require('jsonwebtoken');
const session = require('express-session')
const customer_routes = require('./router/auth_users.js').authenticated;
const genl_routes = require('./router/general.js').general;

const app = express();

app.use(express.json());

app.use("/customer",session({secret:"fingerprint_customer",resave: true, saveUninitialized: true}))

app.use("/customer/auth/*", function auth(req,res,next){
//Write the authenication mechanism here
});

const PORT =5000;

app.use("/customer", customer_routes);
app.use("/", genl_routes);

app.listen(PORT,()=>console.log("Server is running"));
```

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2. The packages required for this lab are defined in as dependencies in `packages.json` as below:

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```

```
  "dependencies": {
    "express": "^4.18.1",
    "express-session": "^1.17.3",
    "jsonwebtoken": "^8.5.1",
    "nodemon": "^2.0.19"
}
```

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Understanding the user routes

Navigate to the `router` directory having the below 3 files:

1. `booksdb.js` - This contains the the preloaded book information for this application.
2. `general.js` - This contains the skeletal implementations for the routes which a general user can access.
3. `auth_users.js` - This contains the skeletal implementations for the routes which an authorized user can access.

Updating the code for the authentication mechanism:

- Navigate to `index.js` and update the authentication code under `app.use("/customer/auth/*", function auth(req,res,next){`:

Hint: Use the session authorization feature (implemented in the Practice project lab) to authenticate a user based on the access token.

Update and test the general user routes in `general.js`.

Task 1:

- Complete the code for getting the list of books available in the shop under `public_users.get('/',function (req, res) {`.

Hint: Use the JSON.stringify method for displaying the output neatly.

- Run `npm install` for installing the required modules & start the server.
- Test the output on Postman.
- Please take a screenshot of the same and save it with the name `1-getallbooks.png` for submitting under Task 1 for the Peer Review Assignment.

Task 2:

- Complete the code for getting the book details based on ISBN under `public_users.get('/isbn/:isbn',function (req, res) {`.

Hint: Retrieve the ISBN from the request parameters

- Test the output on Postman.
- Please take a screenshot of the same and save it with the name `2-gedetailsISBN.png` for submitting under Task 2 for the Peer Review Assignment.

Task 3:

- Complete the code for getting the book details based on the author under `public_users.get('/author/:author',function (req, res) {`.

Hints:
1. Obtain all the keys for the 'books' object.
2. Iterate through the 'books' array & check the author matches the one provided in the request parameters.

- Test the output on Postman.
- Please take a screenshot of the same and save it with the name `3-getbooksbyauthor.png` for submitting under Task 3 for the Peer Review Assignment.

Task 4:

- Complete the code for getting the book details based on the title under `public_users.get('/title/:title',function (req, res) {`.

Hint: This will be similar to Exercise 3

- Test the output on Postman.
- Please take a screenshot of the same and save it with the name `4-getbooksbytitle.png` for submitting under Task 4 for the Peer Review Assignment.

Task 5:

- Complete the code for getting book reviews under `public_users.get('/review/:isbn',function (req, res) {`.

Hint: Get the book reviews based on ISBN provided in the request parameters.

- Please take a screenshot of the same and save it with the name `5-getbookreview.png` for submitting under Task 5 for the Peer Review Assignment.

Task 6:

- Complete the code for registering a new user

Hint: The code should take the 'username' and 'password' provided in the body of the request for registration. If the username already exists, it must mention the same & must also show other errors like eg. when username &/ password are not provided.

- Test the output on Postman.
- Please take a screenshot of the same and save it with the name `6-register.png` for submitting under Task 6 for the Peer Review Assignment.

Update and test the authenticated user routes in `auth_users.js`.

Task 7:

- Complete the code for logging in as a registered user.

Hint: The code must validate and sign in a customer based on the username and password created in Exercise 6. It must also save the user credentials for the session as a JWT.
As you are required to login as a customer, while testing the output on Postman, use the endpoint as "customer/login"

- Test the output on Postman.
- Please take a screenshot of the same and save it with the name `7-login.png` for submitting under Task 7 for the Peer Review Assignment.

Task 8:

- Complete the code for adding or modifying a book review.

Hint: You have to give a review as a request query & it must get posted with the username (stored in the session) posted. If the same user posts a different review on the same ISBN, it should modify the existing review. If another user logs in and posts a review on the same ISBN, it will get added as a different review under the same ISBN.

- Test the output on Postman.
- Please take a screenshot of the same and save it with the name `8-reviewadded.png` for submitting under Task 8 for the Peer Review Assignment.

Task 9:

Complete the code for deleting a book review under `regd_users.delete("/auth/review/:isbn", (req, res) => {`

Hint: Filter & delete the reviews based on the session username, so that a user can delete only his/her reviews and not other users' .

- Test the output on Postman.
- Please take a screenshot of the same and save it with the name `9-deletereview.png` for submitting under Task 9 for the Peer Review Assignment.

With this you have implemented and tested the codes for the general and authenticated user routes.

Improving the scope of Tasks 1-4 using Promises or Async-Await

You will now use Promise callbacks or Async-Await functions for doing the same functionality which we covered synchronously in Tasks 1-4.

Task 10:

- Add the code for getting the list of books available in the shop (done in Task 1) using Promise callbacks or async-await with Axios.

Hint: Refer to this lab on Promises and Callbacks.

- Please ensure that the `general.js` file has the code for getting the list of books available in the shop using Promise callbacks or async-await with Axios is covered. This will be used for the evaluation of Task 10.

Task 11:

- Add the code for getting the book details based on ISBN (done in Task 2) using Promise callbacks or async-await with Axios.

Hint: Refer to this lab on Promises and Callbacks.

- Please ensure that the `general.js` file has the code for getting the book details based on ISBN using Promise callbacks or async-await with Axios is covered. This will be used for the evaluation of Task 11.

Task 12:

- Add the code for getting the book details based on Author (done in Task 3) using Promise callbacks or async-await with Axios.

Hint: Refer to this lab on Promises and Callbacks.

- Please ensure that the `general.js` file has the code for or getting the book details based on Author using Promise callbacks or async-await with Axios is covered. This will be used for the evaluation of Task 12.

Task 13:

- Add the code for getting the book details based on Title (done in Task 4) using Promise callbacks or async-await with Axios.

Hint: Refer to this lab on Promises and Callbacks.

- Please ensure that the `general.js` file has the code for or getting the book details based on Title using Promise callbacks or async-await with Axios is covered. This will be used for the evaluation of Task 13.

Github repo updation for peer review submission

Task 14:

- Please commit and push all the changes to your forked Github repo.

  Note: Please refer to this lab, if you need any help in committing and pushing changes to your repo.

- Your Github repo link will be used for the evaluation of Task 14 in the peer review assignment.

Congratulations! You have completed the Final project!

Summary:

In this lab, you have built a server-side online book review application, integrated it with a secure REST API using JWT based session level authentication, and tested the built application using Promises callbacks or Async-Await functions.

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Changelog

Date	Version	Changed by	Change Description
19-09-2022	1.0	K Sundararajan	Initial version created
20-10-2022	1.1	K Sundararajan	Updated instructions
18-11-2022	1.2	K Sundararajan	Updated instructions based on Coursera Beta testing feedback
25-11-2022	1.3	K Sundararajan	Updated instructions based on edX Beta testing feedback
29-11-2022	1.4	K Sundararajan	Spelling corrections based on Beta testing feedback
10-01-2023	1.5	Lavanya Rajalingam	Grammar checks

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